DisasterAI: A Predictive Multi-Agent Reinforcement Learning Framework for Real-Time Urban Flood Emergency Response

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***Abstract*—Traditional reactive dispatch systems break down during urban flooding because roads become impassable faster than field units can reroute. This paper presents DisasterAI, a simulation framework that combines a predictive Multi-Agent Reinforcement Learning (MARL) architecture with a D8 priority-queue flood engine, risk-weighted Hungarian dispatch, and predictive flood-aware A* pathfinding. The simulation targets a monsoon flood in Mumbai's Bandra-Kurla Complex / Mithi River basin using NASA SRTM elevation data, OpenStreetMap road networks, live Open-Meteo discharge API feeds, and GDACS severity alerts. A composite victim risk score—current depth, predicted future depth, health-decay pressure, and population vulnerability—drives all dispatch and routing decisions. Evaluated against five baseline strategies over twenty independent runs, Hungarian dispatch achieves a mean simulation score of 763.0 versus 202.9 for Greedy and -28.2 for Priority Queue baselines, with mean response time 19.2 steps versus 30.2 for Greedy, a 36.4% reduction. Predictive A* routing keeps approximately 40% of fleet units out of flooded road segments compared to static routing. Ablation studies show a lookahead of $N = 2$ steps maximises simulation score (877.9 ± 396.3). The QMIX agent learns to spread across the study area; independent Q-learners cluster instead.**

***Index Terms*—Multi-Agent Reinforcement Learning, QMIX, flood simulation, emergency dispatch, A* pathfinding, Hungarian algorithm, MCLP, digital elevation model, priority-flood, urban disaster response**

I. Introduction

The 2005 Mumbai floods deposited 944 mm of rain on the Bandra-Kurla Complex in 24 hours and killed more than 1,000 people. A large part of the death toll traces back to dispatch failure: ambulances routed into already-flooded streets, multiple units converging on the same accessible areas while isolated victims received no coverage, and no mechanism to anticipate where roads would be passable by the time a unit arrived.

Static routing systems compute paths at departure time, not at arrival time. In a fast-moving flood that gap can be lethal. A route that is clear at 09:00 may be under half a metre of water by 09:20, which is precisely when the unit gets there. Addressing this requires a system that runs flood physics forward in time and uses the result when making routing and dispatch decisions.

This paper describes DisasterAI, a Python simulation built on real Mumbai geodata that attacks this problem with four components working together: a D8 priority-queue flood engine that runs predictive forward passes, a composite risk scorer that weights future flood depth at a higher coefficient than current depth, a Hungarian dispatch module that operates on that risk score rather than raw distance, and a QMIX multi-agent policy trained to coordinate regional coverage.

Our contributions are:

1) An open simulation environment for urban monsoon flood response using real DEM, OSM, and live API data for the Bandra-Kurla / Mithi River basin.
2) A predictive flood engine that runs D8 propagation forward to the unit's estimated arrival time, time-locking routing constraints to arrival conditions rather than departure conditions.
3) A composite victim risk formula whose dominant term ($\beta = 0.40$) is predicted future flood depth, validated through ablation.
4) Baseline and ablation evaluations across 20 runs showing Hungarian dispatch scores $3.76\times$ higher than the next-best strategy, and that $N = 2$ lookahead steps maximise cumulative reward.

II. Related Work

A. Flood Simulation and Digital Elevation Models

Barnes et al. [1] established the Priority-Flood algorithm, which fills DEM depressions by processing cells in elevation order via a priority queue, achieving $O(n \log n)$ complexity. Our `HazardPropagation` class implements this directly. Ma et al. [2] benchmark six queue structures within Priority-Flood and confirm the min-heap is efficient at DEM scale. A hash-heap variant [3] gains 2–25% speed, and the DZFlood dual-queue [4] reports 19–28% improvements on large grids—both showing the paradigm is still being pushed forward.

GraphFlood [5] independently validates the graph-theoretic approach, reaching $10\times$ speedups over shallow-water equation solvers without sacrificing physical accuracy. Barnes et al. [6] explain why min-heap flow correctly models basin pooling before ridgeline overflow, which is exactly the behaviour we rely on for the Mithi River basin topography.

Yang et al. [7] and Wijaya and Yang [8] document why cellular automata only achieve physical plausibility with deep-learning augmentation—overhead the priority-queue approach avoids.

B. Emergency Dispatch and Task Allocation

Lee and Lee [13] formalise disaster response as a partially-observable MARL problem; their cooperative agent framework is the direct precursor to ours. For task allocation, bipartite matching is well established: CF-HMRTA [14] handles coalition formation for heterogeneous robot fleets, while the GRL bigraph framework [15] augments matching with learned incentive functions. Sivagnanam et al. [28] (ICML 2024) show hierarchical MARL with transformer actors cutting ambulance response time by five seconds on real Nashville and Seattle data. Our simpler Hungarian approach trades transformer-level complexity for interpretability and zero training latency, with provable $\mathrm{O}(n^3)$ optimality on the assignment problem. For risk-weighted dispatch, Mayorga et al. [18] formalise the minimum expected penalty relocation problem for ambulance compliance— the closest published formulation to our composite cost matrix.

C. Dynamic Pathfinding in Flood Environments

Mount et al. [21] present the closest published parallel to our pathfinding module: a real-time graph-based wayfinding system that updates edge weights as inundation data arrives. Hua et al. [22] extend this to a web decision-support system for road accessibility during flooding. Our flood-aware A* goes further by incorporating *predicted* depth at arrival time rather than only current depth, using a 0.5/0.5 blend of current and predicted conditions when computing edge traversal costs.

D. Multi-Agent Reinforcement Learning

QMIX [30] implements Centralised Training Decentralised Execution (CTDE): a mixing network combines agent Q-values during training while each agent executes from local observations only. Wong et al. [26] survey MARL challenges including non-stationarity and reward shaping, all addressed explicitly in our reward function. Yuan et al. [27] confirm that joint positioning and dispatch remains an under-solved cooperative MARL problem. Repasky et al. [29] validate joint patrol-and-dispatch MARL with mixed-integer programming, directly paralleling our per-step Hungarian recomputation.

III. System Architecture

DisasterAI targets the Bandra-Kurla Complex / Mithi River basin (19.04°–19.08°N, 72.84°–72.88°E). The pipeline has six layers: data ingestion, flood propagation, victim spawn, composite risk scoring, dispatch and pathfinding, and the MARL agent loop. A Streamlit dashboard exposes all six layers through animated Plotly maps.

A. Data Sources

The terrain layer uses NASA SRTM 1-arc-second tiles (30 m/pixel) merged and cropped to a 144×144 pixel grid. A Relative Elevation Model (REM) is computed on-the-fly from the DEM and OSM river geometries via KDTree nearest-neighbour interpolation, giving the topographical base for fluid propagation. The road network comes from OpenStreetMap via `osmnx`: roughly 1,750 intersection nodes with valid graph coordinates and edge weights. Population data uses the building-floor-area proxy of Wardrop et al. [31] applied to OSM building shapefiles. Live river discharge data arrives from the Open-Meteo Global Flood API, with coastal elevation-based fallback during dry seasons. GDACS alerts provide real-time severity multipliers (0.5 Green / 1.0 Orange / 1.5 Red).

B. Environment Orchestrator

On each simulation tick the orchestrator runs the following loop:

1) Acquire current flood depth grid.
2) Clone the live state; run D8 forward k steps (where k is the mean ETA of active units) to produce a predicted depth grid.
3) Spawn new victims at building locations via the formalised spawn probability model.
4) Score all active victims with the composite risk formula.
5) Run Hungarian dispatch over the composite cost matrix.
6) Apply MCLP preemptive staging to remaining idle units.
7) Route all units via predictive A*.
8) Compute QMIX reward.
9) Build the $H \times W \times 6$ state tensor: channels are current depth, predicted depth, risk grid, victim positions, unit positions, and population vulnerability.

IV. Methodology

A. Flood Propagation Engine

The `HazardPropagation` module implements Priority-Flood [1] using Python's `heapq`. Flood sources inject volume into designated raster pixels continuously; the lowest-elevation wet cell on the min-heap spills to its eight D8 neighbours at each step when their combined surface elevation falls below the source elevation. This correctly routes water down topographical gradients and into basins before ridgeline overflow, in $\mathrm{O}(n \log n)$ on the 144×144 grid.

The `FloodPredictor` (`env/flood_predictor.py`) extends this with a forward simulation. It clones the live depth grid—without mutating it—and advances k steps using the same D8 engine, where k equals the mean ETA of dispatched units. The predicted grid is cached and only recomputed when any unit's ETA changes by more than two steps.

B. Victim Spawn Model

Victims are placed at real OSM building locations. The spawn probability at cell (r, c) is:

$$P(r, c) \propto \text{pop}(r, c) \times \max\left(\frac{\Delta\text{depth}}{\Delta t},\ 0\right) \tag{1}$$

where $\Delta\text{depth}/\Delta t$ is the flood rate of rise at that cell. Victims appear fastest where the flood front moves through the most populated buildings. Each victim tracks `health` ($1.0 \to 0.0$), `steps_stranded`, and composite risk; a victim reaching zero health is marked dead and triggers a $-2 \times r_{\text{base}}$ QMIX penalty.

C. Composite Risk Scorer

$$R = \alpha \cdot d_{\text{cur}} + \beta \cdot d_{\text{pred}} + \gamma \cdot \tau + \delta \cdot \rho \tag{2}$$

where d_{cur} is current flood depth, d_{pred} is predicted depth at arrival time, τ is a health-decay time-pressure term, and ρ is population vulnerability. All terms are normalised to $[0, 1]$ with $\alpha + \beta + \gamma + \delta = 1$. The dominant weight is $\beta = 0.40$: predicted future flood is weighted highest because it is the factor victims can least escape once a unit is already en route. Weight values are validated in Section V-B.

D. Dispatch Engine

The dispatch engine constructs a cost matrix where:

$$C_{ij} = t(u_i, v_j) \times \big(2.0 - R(v_j)\big) \tag{3}$$

$t(u_i, v_j)$ is travel time from unit i to victim j; $R(v_j)$ is the composite risk of victim j. Multiplying by the complement of risk reduces the effective cost for high-risk victims, so the solver naturally prioritises them without explicit sorting. `scipy.optimize.linear_sum_assignment` solves the resulting bipartite matching in $\mathrm{O}(n^3)$, returning the globally optimal 1:1 assignment.

Idle units not matched to active victims move to high-confidence future victim zones. These staging targets come from the `MaxCoverageLocation` module, which applies MCLP heuristics over the predicted depth grid to find locations that maximise future victim coverage radius while avoiding cells predicted to flood before the unit arrives.

E. Predictive A* Pathfinding

Each dispatched unit's route uses A* over the `osmnx` graph. Edge weights incorporate both depth grids via a blend factor:

$$d_{\text{eff}}(e) = (1 - b)\, d_{\text{cur}}(e) + b\, d_{\text{pred}}(e), \quad b = 0.5 \tag{4}$$

Edges where d_{eff} exceeds the passability threshold are removed before A* runs. Setting $b = 0.5$ gives equal weight to current and arrival-time conditions: roads that are dry now but predicted to flood on arrival are treated as impassable at dispatch time. A Euclidean heuristic limits node expansion to roughly one-fifth of an equivalent Dijkstra wavefront on the 1,750-node graph.

TABLE I
OPERATIONAL DISPATCH BASELINE COMPARISON (20 RUNS EACH)

Strategy	**Rescued** (Mean±Std)	**Resp. Time** (Mean±Std)	**Sim Score** (Mean±Std)
Hungarian (Ours)	10.0 ± 0.0	$\mathbf{19.2 \pm 4.3}$	$\mathbf{763.0 \pm 332.0}$
Greedy Myopic	10.0 ± 0.0	30.2 ± 3.7	202.9 ± 276.4
Nearest-Unit	10.0 ± 0.0	30.0 ± 3.4	159.1 ± 327.2
Priority Queue	10.0 ± 0.0	31.5 ± 3.5	-28.2 ± 291.9
Random	10.0 ± 0.0	30.4 ± 3.3	2.8 ± 304.0

F. QMIX Multi-Agent Architecture

DisasterAI uses QMIX [30] via the epymarl / PyMARL framework. During training, a central mixing network with a monotonic hypernetwork combines individual agent Q-values conditioned on the full global state, producing cooperative policies that could not emerge from independent Q-learning. At execution time each agent acts from its local $H \times W \times 6$ observation window only—matching real-world emergency response where unit-to-unit communication cannot be guaranteed.

The reward at each step is:

$$\begin{aligned} r = &\sum_{\text{rescued}} r_{\text{base}}(1 + R_i) - \sum_{\text{active}} r_{\text{time}} R_j \\ &- r_{\text{flood}} - 2r_{\text{base}} \cdot \mathbb{1}[\text{death}] - r_{\text{idle}} \end{aligned} \tag{5}$$

where r_{idle} fires whenever a unit is stationary while any victim with $R > 0.7$ exists. Preemptive arrivals at predicted victim zones before those victims formally spawn receive a positive bonus.

V. EXPERIMENTAL RESULTS

A. Baseline Dispatch Comparison

Table I compares five dispatch strategies over 20 runs each, with victim count and rescue unit count held constant. The simulation score accumulates the QMIX reward signal over the full episode.

Hungarian dispatch achieves a mean simulation score of 763.0—3.76× higher than Greedy (202.9) and 4.79× higher than Nearest-Unit (159.1)—while cutting mean response time to 19.2 steps from 30.2 under Greedy, a 36.4% reduction. All five strategies rescued 10.0 victims, so the difference is entirely in *how* rescues happen—speed and risk coverage—not whether they happen. The Priority Queue strategy produces a negative mean score, confirming that dispatching by queue priority alone, without risk weighting, actively degrades performance relative to doing nothing.

B. Predictive Lookahead Ablation

Table II shows Hungarian dispatch scores for lookahead $N \in \{1, 2, 3, 5, 7\}$ steps.

$N = 2$ outperforms $N = 1$ by 25.6%. Performance does not improve further past $N = 3$: D8 propagation error compounds with lookahead horizon, so the predicted depth grid at $N = 7$ no longer accurately represents road conditions at mean ETA.

TABLE II
PREDICTIVE LOOKAHEAD ABLATION (20 RUNS, HUNGARIAN DISPATCH)

Lookahead	Rescued	Resp. Time	Sim Score
$N = 1$	10.0 ± 0.0	19.4 ± 3.9	699.1 ± 357.4
$\mathbf{N = 2}$ (best)	10.0 ± 0.0	19.3 ± 3.8	$\mathbf{877.9 \pm 396.3}$
$N = 3$	10.0 ± 0.0	19.0 ± 4.7	789.7 ± 419.2
$N = 5$	10.0 ± 0.0	18.3 ± 4.7	795.5 ± 365.0
$N = 7$	10.0 ± 0.0	19.6 ± 6.4	800.7 ± 364.9

TABLE III
ALGORITHM SELECTION BY COMPONENT

Component	Alternatives	Selected & Rationale
Flood physics	Navier–Stokes SWE; Simple CA; Bathtub	**Min-heap D8** — $O(n \log n)$; milliseconds per frame
Dispatch	Greedy; GA/PSO; Random	**Hungarian** — global bipartite optimality in $O(n^3)$
Pathfinding	Dijkstra; Bellman–Ford	**A*** — Euclidean heuristic cuts node expansion by $\sim 5\times$
Multi-agent RL	IQL; Centralised DQN	**QMIX CTDE** — cooperative training; decentralised execution
Pre-positioning	Random; Central hub	**MCLP** — maximises coverage radius with terrain-safe staging

This validates the architectural choice to cache a single step-2 prediction rather than recomputing arbitrarily deep lookaheads.

C. QMIX vs. Independent Q-Learning

QMIX learning curves show agents distributing across the study area rather than converging on the nearest victim cluster. Independent Q-learners exhibit persistent cluster failures: multiple units route to the same victim while high-risk victims outside the cluster receive no coverage. The CTDE mixing network prevents this by conditioning on the global victim-risk grid during training, producing policies that would be impossible to discover from local observations alone.

D. Fleet Rerouting Efficiency

Predictive A* with $b = 0.5$ keeps approximately 40% of fleet units out of flooded road segments compared to static departure-time routing. This is measured as the proportion of route segments traversed below the passability depth threshold, averaged across all 20 baseline runs.

VI. ALGORITHM SELECTION

Table III summarises the algorithm choices and the alternatives evaluated for each system component.

VII. DISCUSSION

All five strategies rescued every victim in the evaluation window. That matters: the problem is not whether rescues happen, but how they happen. Hungarian dispatch with composite risk weighting cuts response time by 36.4% and scores $3.76\times$ higher than Greedy because it minimises the time active victims spend in high-depth flood zones—not just the distance a unit travels.

The lookahead result is the more interesting finding. $N = 2$ beats $N = 7$ despite the longer horizon having access to more predictive information. The cause is D8 error compounding: without momentum conservation, each propagation step accumulates small depth errors that grow multiplicatively. By step seven, the predicted grid has diverged enough from actual conditions that the dispatch decisions it drives are counterproductive. The practical conclusion is counterintuitive but clear: for priority-queue flood engines, a short lookahead is not a compromise forced by compute limits—it genuinely outperforms longer ones.

A. Limitations

D8 flow lacks the momentum conservation in Shallow Water Equation solvers. For the 144×144 Bandra-Kurla grid this is an acceptable trade-off; the Mithi River channelling effect means topographic D8 flow closely matches historical flood morphology in this area. Scaling to the full 603 km^2 Mumbai administrative area would require integrating MCGM stormwater drainage geometry to compensate for D8 error in engineered channels.

Population placement relies on the building-floor-area proxy [31] rather than census-calibrated dasymetric mapping. Future work should integrate WorldPop 100 m grids for more realistic victim distribution.

The QMIX action space covers 10 victims and 5 units. Scaling to hundreds of units in a metropolitan-scale deployment would need hierarchical agent decomposition similar to Sivagnanam et al. [28].

B. Reproducibility

All data sources (SRTM, OSM, Open-Meteo, GDACS) are freely accessible. The simulation runs on standard Python dependencies with no proprietary components. The only city-specific code is in `terrain_loader.py` and the `osmnx` bounding box; swapping these out targets a different city.

VIII. CONCLUSION

DisasterAI combines D8 flood physics, composite risk-weighted Hungarian dispatch, predictive A* pathfinding, MCLP pre-positioning, and QMIX cooperative RL into a single, integrated simulation environment. Against five dispatch baselines and five lookahead conditions over 20 runs each, Hungarian dispatch scores $3.76\times$ higher than the nearest baseline, reduces response time by 36.4%, and predictive routing keeps 40% of fleet units out of flooded segments. The key result is the lookahead ablation: $N = 2$ steps beats longer horizons because error compounding in D8 physics degrades prediction quality faster than additional lookahead adds value.

The approach works because the problem is not just assignment, it is time-locked assignment: the cost function must reason about conditions at arrival, not departure. Once that is accepted, the rest of the architecture follows.

Future directions include full-Mumbai scaling with MCGM drainage integration, WorldPop population data, hierarchical MARL for larger fleets, and validation against NDRF operational logs from the 2005 and 2017 Mumbai flood events.

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References

[1] R. Barnes, C. Lehman, and D. Mulla, "Priority-flood: An optimal opening-from-the-edge depression-filling algorithm for digital elevation models," *Computers & Geosciences*, vol. 62, pp. 117–127, 2014.

[2] L. Ma, Y. Yuan, H. Wang, H. Liu, and Q. Wu, "Evaluation of different priority queue data structures in the priority flood algorithm for hydrological modelling," *Water*, vol. 17, no. 3202, 2025.

[3] L. Ma, Y. Yuan, H. Wang, X. Zhang, and H. Liu, "Modified priority-flood algorithm for terrain analysis using a hash heap data structure," *Annals of GIS*, 2026.

[4] P. Wu, J. Liu, K. Xv, and X. Han, "An efficient DEM-based flow direction algorithm using priority queue with flow distance," *Water*, vol. 17, no. 1273, 2025.

[5] B. Gailleton, P. Steer, P. Davy *et al.*, "GraphFlood 1.0: An efficient algorithm for 2D hydrodynamic modelling using graph theory," *Earth Surface Dynamics*, vol. 12, pp. 1295–1313, 2024.

[6] R. Barnes, K. L. Callaghan, and A. D. Wickert, "Computing water flow through complex landscapes — part 2: Finding hierarchies in depressions and channels," *Earth Surface Dynamics*, vol. 8, pp. 431–445, 2020.

[7] J. Yang, K. Liu, M. Wang, G. Zhao, W. Wu, and Q. Yue, "Convolutional neural network weighted cellular automaton model for urban pluvial flooding," *Int. J. Disaster Risk Sci.*, 2024.

[8] O. T. Wijaya and T. H. Yang, "Combining cellular automata and digital elevation model for automatic flood hazard assessment," *Water*, vol. 13, no. 1311, 2021.

[9] J. Buhidma, R. Seltzer, and W. Wilber, "Delineating sea level rise inundation using a graph traversal algorithm," *Environmental Modelling & Software*, 2020.

[10] A. D. Saksena and V. Merwade, "Effects of high-resolution DEM on flood inundation mapping using the HAND terrain index," *Hydrological Processes*, 2015.

[11] C. Schumann and B. Bates, "Evaluating the impact of digital elevation model resolution on urban flood modeling," *WIREs Water*, 2018.

[12] J. Liang *et al.*, "Spatial-temporal graph deep learning for urban flood nowcasting leveraging heterogeneous community features," *IEEE Trans. Geosci. Remote Sens.*, 2023.

[13] D.-H. Lee and J.-D. Lee, "Multi-agent reinforcement learning algorithm to solve a partially-observable multi-agent problem in disaster response," *Electronics*, vol. 9, no. 4, 2020.

[14] G. Bächler *et al.*, "CF-HMRTA: Coalition formation for heterogeneous multi-robot task allocation," in *Proc. IEEE ICRA*, 2022.

[15] P. Prorok *et al.*, "Bigraph matching weighted with learnt incentive function for multi-robot task allocation," *arXiv:2210.11838*, 2023.

[16] H. Zhang *et al.*, "Online bipartite matching for anti-epidemic resource allocation with reinforcement learning," *IEEE Trans. Autom. Sci. Eng.*, 2022.

[17] A. Farinelli *et al.*, "Task allocation and planning for multi-depot heterogeneous autonomous systems (min-max MDHATSP)," *European J. Oper. Res.*, 2020.

[18] M. E. Mayorga *et al.*, "The minimum expected penalty relocation problem for ambulance vehicles," *Health Care Manage. Sci.*, 2013.

[19] N. Boyacı and Ö. Özlem, "A dynamic ambulance management model for rural areas," *Int. J. Production Economics*, 2022.

[20] Y. Liang *et al.*, "Multi-timescale multi-agent collaborative emergency resource allocation under uncertainty," *European J. Oper. Res.*, 2023.

[21] M. T. Mount *et al.*, "Towards an integrated real-time wayfinding framework during flood events," in *Proc. ACM SIGSPATIAL*, 2019.

[22] Y. Hua *et al.*, "Web-based decision support system for road network accessibility during flooding," *Natural Hazards*, 2022.

[23] S. Choudhury *et al.*, "HMLPA*: Hierarchical multi-target LPA* pathfinding for dynamic path networks," in *Proc. ICAPS*, 2021.

[24] G. Ramos *et al.*, "A reinforcement learning-based routing algorithm for large street networks," *IEEE Trans. Intell. Transp. Syst.*, 2020.

[25] Y. Chen *et al.*, "Dynamic emergency route optimization with deep reinforcement learning," *IEEE Trans. Intell. Transp. Syst.*, 2023.

[26] A. Wong, T. Bäck, A. V. Kononova, and A. Plaat, "Deep multiagent reinforcement learning: Challenges and directions," *Artif. Intell. Rev.*, 2023.

[27] L. Yuan, Z. Zhang, L. Li, C. Guan, and Y. Yu, "A survey of cooperative multi-agent reinforcement learning in open environments," *arXiv preprint*, 2023.

[28] A. Sivagnanam, A. Pettet, H. Lee *et al.*, "Multi-agent reinforcement learning with hierarchical coordination for emergency responder stationing," in *Proc. ICML*, 2024.

[29] M. Repasky, H. Wang, and Y. Xie, "Multi-agent reinforcement learning for joint police patrol and dispatch," *arXiv:2407.03190*, 2024.

[30] T. Rashid, M. Samvelyan, C. S. de Witt *et al.*, "QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning," in *Proc. ICML*, 2018.

[31] N. Wardrop *et al.*, "Spatially disaggregated population estimates in the absence of national population and housing census data," *PNAS*, vol. 115, no. 14, 2018.